1.

$$F(\vec{v},t) = \overbrace{n_1 \left(\frac{m}{2\pi kT_1}\right)^{3/2} \exp\left\{-\frac{m}{2kT_1}\left[(V_x-u_{1x})^2+(V_y-u_{1y})^2+(V_z-u_{1z})^2\right]\right\}}^{F_1}$$
$$+ \underbrace{n_2 \left(\frac{m}{2\pi kT_2}\right)^{3/2} \exp\left\{-\frac{m}{2kT_2}\left[(V_x-u_{2x})^2+(V_y-u_{2y})^2+(V_z-u_{2z})^2\right]\right\}}_{F_2}$$

a) $\rho = \langle mF \rangle = m\langle F \rangle = m\langle F_1 + F_2 \rangle = m\langle F_1 \rangle + m\langle F_2 \rangle$

$$\langle F_1 \rangle = \iiint_{\infty} d\vec{V}\; n_1 \left(\frac{m}{2\pi kT_1}\right)^{3/2} \exp\left\{-\frac{m}{2kT_1}\left[(V_x-u_{1x})^2+(V_y-u_{1y})^2+(V_z-u_{1z})^2\right]\right\}$$

$$= n_1 \left(\frac{m}{2\pi kT_1}\right)^{3/2} \iiint_{\infty} \exp\left\{-\frac{m}{2kT_1}\left[(V_x-u_{1x})^2+(V_y-u_{1y})^2+(V_z-u_{1z})^2\right]\right\}$$

$$= n_1 \left(\frac{m}{2\pi kT_1}\right)^{3/2} \int_{\infty} \exp\left\{-\frac{m}{2kT_1}(V_x-u_{1x})^2\right\} dV_x \int_{\infty} \exp\left\{-\frac{m}{2kT_1}(V_y-u_{1y})^2\right\} dV_y \int_{\infty} \exp\left\{-\frac{m}{2kT_1}(V_z-u_{1z})^2\right\} dV_z$$

define $C_i = V_i - u_{1i}$, then perform substitution within the interval.

$$= n_1 \left(\frac{m}{2\pi kT_1}\right)^{3/2} \underbrace{\int_{\infty} \exp\left\{-\frac{m}{2kT_1}C_x^2\right\} dC_x}_{I} \underbrace{\int_{\infty} \exp\left\{-\frac{m}{2kT_1}C_y^2\right\} dC_y}_{I} \underbrace{\int_{\infty} \exp\left\{-\frac{m}{2kT_1}C_z^2\right\} dC_z}_{I}$$

the integral I is of known form $\int_{-\infty}^{\infty} e^{-\beta t^2} dt = \sqrt{\frac{\pi}{\beta}}$,

so for $\beta = \frac{m}{2kT_1}$, $\int_{-\infty}^{\infty} e^{-\beta t^2} dt = \sqrt{\frac{2kT_1\pi}{m}}$

$$= n_1 \left(\frac{m}{2\pi kT_1}\right)^{3/2} \left(\sqrt{\frac{2kT_1\pi}{m}}\right)^3 = n_1$$, which is what we expect from the notes.

by only altering indices, we can perform the same calculations on F_2 to get $\langle F_2 \rangle = n_2$

so we have $\rho = m\left(\langle F_1 \rangle + \langle F_2 \rangle\right) = m(n_1+n_2)$

b) $\rho\vec{u} = m\langle \vec{v} F \rangle = m\langle V_x F \rangle \hat{x} + m\langle V_y F \rangle \hat{y} + m\langle V_z F \rangle \hat{z}$

begin with $\langle V_x F \rangle$, we can perform the same separation of F

$\langle V_x F \rangle = \langle V_x F_1 \rangle + \langle V_x F_2 \rangle$, so just evaluate $\langle V_x F_1 \rangle$

$$\langle V_x F_1 \rangle = n_1 \left(\frac{m}{2\pi k T_1}\right)^{3/2} \iiint_{\infty} V_x \exp\left\{-\frac{m}{2kT_1}\left[(V_x-u_{1x})^2+(V_y-u_{1y})^2+(V_z-u_{1z})^2\right]\right\} d\vec{v}$$

we can substitute $C_i = V_i - u_{1i}$, and $V_i = C_i + u_{1i}$ to get...

$$\langle V_x F_1 \rangle = n_1 \left(\frac{m}{2\pi k T_1}\right)^{3/2} \iiint_{\infty} (C_x+u_{1x}) \exp\left\{-\frac{m}{2kT_1}\left[C_x^2+C_y^2+C_z^2\right]\right\} d\vec{C}$$

$$= n_1 \left(\frac{m}{2\pi k T_1}\right)^{3/2} \Bigg(u_{1x} \underbrace{\iiint_{\infty} \exp\left\{-\frac{m}{2kT_1}\left[C_x^2+C_y^2+C_z^2\right]\right\} d\vec{C}}_{\text{result from a) } = (2kT_1\pi/m)^{3/2}} + \underbrace{\iiint_{\infty} C_x \exp\left\{-\frac{m}{2kT_1}\left[C_x^2+C_y^2+C_z^2\right]\right\} d\vec{C}}_{=0 \text{ (odd in } x)} \Bigg)$$

$$= n_1 \left(\frac{m}{2\pi k T_1}\right)^{3/2} u_{1x} \left(\frac{2\pi k T_1}{m}\right)^{3/2}$$

$\langle V_x F_1 \rangle = n_1 u_{1x}$, and since F_1 is isotropic and seperable we can see that just changing indices gives results in the y and z directions.

$\langle V_y F_1 \rangle = n_1 u_{1y}$, $\langle V_z F_1 \rangle = n_1 u_{1z}$

then using the same argument as in a, we can just change indices from 1 to 2 to get results of F_2 from F_1 results.

$\langle V_x F_2 \rangle = n_2 u_{2x}$, $\langle V_y F_2 \rangle = n_2 u_{2y}$, $\langle V_z F_2 \rangle = n_2 u_{2z}$

now all of these results form our solution

$\langle \vec{V} F \rangle = \langle V_x F \rangle \hat{x} + \langle V_y F \rangle \hat{y} + \langle V_z F \rangle \hat{z}$

$\langle \vec{V} F \rangle = \rho \vec{u} = m(n_1+n_2)\vec{u}$, $\vec{u} = \dfrac{\langle \vec{V} F \rangle}{n_1+n_2}$

$$\vec{u} = \frac{1}{n_1+n_2}\left[(n_1 u_{1x} + n_2 u_{2x})\hat{x} + (n_1 u_{1y} + n_2 u_{2y})\hat{y} + (n_1 u_{1z} + n_2 u_{2z})\hat{z}\right]$$

$$\vec{u} = \frac{n_1 \vec{u}_1 + n_2 \vec{u}_2}{n_1+n_2}$$

$$u^2 = \vec{u}\cdot\vec{u} = (n_1\vec{u}_1 + n_2\vec{u}_2)\cdot(n_1\vec{u}_1 + n_2\vec{u}_2)/(n_1+n_2)^2$$

$$= \left(n_1^2 u_1^2 + 2 n_1 n_2 \vec{u}_1\cdot\vec{u}_2 + n_2^2 u_2^2\right)/(n_1+n_2)^2$$

c) the gas pressure can be obtained from the contracted second moment

$$\frac{m}{2}\langle V^2 \mathcal{F}\rangle = \frac{1}{2}\rho u^2 + \frac{3}{2}p \quad \Rightarrow \quad p = \frac{1}{3}\left(m\langle V^2\mathcal{F}\rangle - \rho u^2\right)$$

so begin by solving $\langle V^2\mathcal{F}\rangle$.

$$\langle V^2\mathcal{F}\rangle = \langle V^2\mathcal{F}_1\rangle + \langle V^2\mathcal{F}_2\rangle$$ where we can then solve $\langle V^2\mathcal{F}_1\rangle$ on its own.

$$\langle V^2\mathcal{F}_1\rangle = n_1\left(\frac{m}{2\pi kT_1}\right)^{3/2}\iiint_{-\infty}^{\infty} V^2 \exp\left\{-\frac{m}{2kT_1}\left[(V_x-u_{1x})^2+(V_y-u_{1y})^2+(V_z-u_{1z})^2\right]\right\} d\vec{V}$$

where we can write... $\vec{C}\cdot\vec{C} = (\vec{V}-\vec{u}_1)\cdot(\vec{V}-\vec{u}_1) = V^2 - 2\vec{u}_1\vec{V} + u_1^2$

so $V^2 = C^2 + 2\vec{u}_1\vec{V} - u_1^2$, where $\vec{V} = \vec{C}+\vec{u}_1$, so $V^2 = C^2 + 2\vec{u}_1\vec{C} + 2u_1^2 - u_1^2$

$$\langle V^2\mathcal{F}_1\rangle = n_1\left(\frac{m}{2\pi kT_1}\right)^{3/2}\iiint_{-\infty}^{\infty}\left(C^2 + 2\vec{u}_1\cdot\vec{V} - u^2\right)\exp\left\{-\frac{m}{2kT_1}\left[C_x^2+C_y^2+C_z^2\right]\right\} d\vec{C}$$

$$= n_1\left(\frac{m}{2\pi kT_1}\right)^{3/2}\Bigg[\underbrace{\iiint_{-\infty}^{\infty} C^2\exp\left\{-\frac{m}{2kT_1}C^2\right\}d\vec{C}}_{①} + \underbrace{\iiint_{-\infty}^{\infty} 2\vec{u}_1\cdot\vec{C}\exp\left\{-\frac{m}{2kT_1}C^2\right\}d\vec{C}}_{②} + \underbrace{\iiint_{-\infty}^{\infty} u_1^2\exp\left\{-\frac{m}{2kT_1}C^2\right\}d\vec{C}}_{③}\Bigg]$$

solve integrals ① - ③ seperately.

$$① = \iiint_{-\infty}^{\infty} C^2\exp\left\{-\frac{m}{2kT_1}C^2\right\}d\vec{C}$$

$$= \Bigg[\underbrace{\int_{-\infty}^{\infty} C_x^2\exp\left(-\frac{m}{2kT_1}C_x^2\right)dC_x}_{I_2}\underbrace{\int_{-\infty}^{\infty}\exp\left(-\frac{m}{2kT_1}C_y^2\right)dC_y}_{I_0}\underbrace{\int_{-\infty}^{\infty}\exp\left(-\frac{m}{2kT_1}C_z^2\right)dC_z}_{I_0}$$

$$+ \underbrace{\int_{-\infty}^{\infty}\exp\left(-\frac{m}{2kT_1}C_x^2\right)dC_x}_{I_0}\underbrace{\int_{-\infty}^{\infty} C_y^2\exp\left(-\frac{m}{2kT_1}C_y^2\right)dC_y}_{I_2}\underbrace{\int_{-\infty}^{\infty}\exp\left(-\frac{m}{2kT_1}C_z^2\right)dC_z}_{I_0}$$

$$+ \underbrace{\int_{-\infty}^{\infty}\exp\left(-\frac{m}{2kT_1}C_x^2\right)dC_x}_{I_0}\underbrace{\int_{-\infty}^{\infty}\exp\left(-\frac{m}{2kT_1}C_y^2\right)dC_y}_{I_0}\underbrace{\int_{-\infty}^{\infty} C_z^2\exp\left(-\frac{m}{2kT_1}C_z^2\right)dC_z}_{I_2}\Bigg]$$

$$= 3\left(I_2 I_0^2\right) \quad \text{where} \quad I_2 = \frac{\sqrt{\pi}}{2\beta^{3/2}} \quad I_0 = \frac{\sqrt{\pi}}{\sqrt{\beta}} \quad \text{for} \quad \beta = \frac{m}{2kT_1}$$

$$= 3\left(\frac{\pi^{3/2}}{2\beta^{5/2}}\right) = 3\left(\frac{\pi^{3/2}}{2}\cdot\left(\frac{2kT_1}{m}\right)^{5/2}\right) = 3\left((2\pi)^{3/2}\left(\frac{kT_1}{m}\right)^{5/2}\right)$$

$$② = \underbrace{\iiint_{-\infty}^{\infty} 2\vec{u}_1\cdot\vec{C}\exp\left\{-\frac{m}{2kT_1}C^2\right\}d\vec{C}}_{=0} = 0$$

(since there are odd terms for each integral)

$$③ = \iiint_{\infty} u_i^2 \exp\left\{-\frac{m}{2kT_1}c^2\right\} d\vec{c} = u_i^2 \iiint_{\infty} \exp\left\{-\frac{m}{2kT_1}c^2\right\} d\vec{c} = u_i^2 \, I_0^3$$

$$= u_i^2 \left(\frac{\pi}{\beta}\right)^{3/2} = u_i^2 \left(\frac{\pi 2kT_1}{m}\right)^{3/2}$$

then plugging ①-③ back in gives.

$$\langle v^2 F_1 \rangle = n_1 \left(\frac{m}{2\pi kT_1}\right)^{3/2} \left[3(2\pi)^{3/2}\left(\frac{2kT_1}{m}\right)^{5/2} + u_i^2 \left(\frac{\pi 2kT_1}{m}\right)^{3/2}\right]$$

$$= 3n_1 kT_1/m + n_1 u_i^2$$

then by just changing indices we get... $\langle v^2 F_2 \rangle = 3n_2 kT_2/m + n_1 u_2^2$

$$\langle v^2 F \rangle = 3n_1 kT_1/m + 3n_2 kT_2/m + n_1 u_1^2 + n_2 u_2^2$$

$$P = \frac{1}{3}\left(m\langle v^2 F\rangle - \rho u^2\right) = \frac{1}{3}\left(3n_1 kT_1 + 3n_2 kT_2 + n_1 m u_1^2 + n_2 m u_2^2 - \rho u^2\right)$$

$$= n_1 kT_1 + n_2 kT_2 + \tfrac{1}{3}\left(n_1 m u_1^2 + n_2 m u_2^2 - (n_1+n_2) m u^2\right)$$

$(n_1+n_2)(n_1 m u_1^2 + n_2 m u_2^2)/(n_1+n_2)$ ④ $\qquad m\,|n_1\vec{u}_1 + n_2\vec{u}_2|^2/(n_1+n_2)$ ⑤

$$④ = \left(n_1^2 m u_1^2 + n_1 n_2 m u_2^2 + n_1 n_2 m u_1^2 + n_2^2 m u_2^2\right)/(n_1+n_2)$$

$$⑤ = \left(n_1^2 m u_1^2 + 2n_1 n_2 u_1 u_2 + n_2^2 m u_2^2\right)/(n_1+n_2)$$

$$④ - ⑤ = \left(n_1 n_2 m u_2^2 + n_1 n_2 m u_1^2 - 2 n_1 n_2 u_1 u_2\right)/(n_1+n_2)$$

$$= \frac{n_1 n_2}{n_1+n_2}|u_1 - u_2|^2$$

$$\boxed{P = n_1 kT_1 + n_2 kT_2 + \frac{1}{3}\frac{m n_1 n_2}{n_1+n_2}|u_1 - u_2|^2}$$

d) If a large number of interparticle collisions occur, the following will occur, then we will reach equilibrium.

→ equilibrium means the system can be described by a single temperature $T = T_1 = T_2$

In addition to this, we know that in the limit of a large number of interparticle collisions, our NDF tends towards the M-B dist.

$$F(\vec{v},t) \sim M(\vec{v}) = n\left(\frac{m}{2\pi kT}\right) \exp\left\{-\frac{m}{2kT}|\vec{v}-\vec{u}|^2\right\}$$

so the new NDF would be the Maxwell-Boltzmann dist, which can be directly recovered from our NDF by setting $n = n_1+n_2$, $T = T_1 = T_2$, and $\vec{u}_1 = \vec{u}_2$

→ for $\vec{u}_1 = \vec{u}_2$, we can calculate the bulk velocity using the result from b) $\vec{u} = \frac{\vec{u}_1 n_1 + \vec{u}_1 n_2}{n_1+n_2} = \vec{u}_1 \frac{(n_1+n_2)}{(n_1+n_2)} = \vec{u}_1 = \vec{u}_2 = (u_x, u_y, u_z)$

→ this gives new equilibrium NDF...

$$F_{equil}(\vec{v},t) = (n_1+n_2)\left(\frac{m}{2\pi kT}\right)^{3/2} \exp\left\{-\frac{m}{2kT}\left[(v_x-u_x)^2+(v_y-u_y)^2+(v_z-u_z)^2\right]\right\}$$

e) for this new distribution $F_{equil}(\vec{v},t)$, this is basically a maxwellian with $n = n_1+n_2$, so the moments for this gas are those of the maxwellian in this case.

density: $V = m \quad \langle mF \rangle = \rho = m\langle F \rangle$

$$\langle F \rangle = (n_1+n_2)\left(\frac{m}{2\pi kT}\right)^{3/2} I_0^3 = (n_1+n_2), \quad \rho = m(n_1+n_2)$$

average (bulk) velocity: $V = m\vec{v} \quad \langle m\vec{v}F \rangle = \rho\vec{u}, \quad \vec{u}_{avg} = \frac{m}{\rho}\langle \vec{v}F \rangle$

$$\langle v_i F \rangle = (n_1+n_2)\left(\frac{m}{2\pi kT}\right)^{3/2} \iiint_{\infty} v_i \exp\left\{-\frac{m}{2kT}(v_x-u_x)^2+(v_y-u_y)^2+(v_z-u_z)^2\right\} d\vec{v}$$

substitute $C_i = v_i - u_i$, so $v_i = C_i + u_i$

$$= (n_1+n_2)\left(\frac{m}{2\pi kT}\right)^{3/2} \iiint_{\infty} (C_i+u_i) \exp\left\{-\frac{m}{2kT}C^2\right\} d\vec{C}, \text{ but all } C_i \text{ int. are odd.}$$

$$= (n_1+n_2)\left(\frac{m}{2\pi kT}\right)^{3/2} u_i \, I_0^3$$

$$= (n_1+n_2)u_i \implies \vec{u}_{avg} = \frac{m}{m(n_1+n_2)}\vec{u}\,(n_1+n_2), \quad \vec{u}_{avg} = \vec{u}$$

Pressure : $V = \frac{1}{2}mv^2$ $\frac{m}{2}\langle v^2 F\rangle = \frac{1}{2}\rho u^2 + \frac{3}{2}p \Rightarrow p = \frac{1}{3}\left(m\langle v^2 F\rangle - \rho u^2\right)$

$$\langle v^2 F\rangle = (n_1+n_2)\left(\frac{m}{2\pi kT}\right)^{3/2}\iiint_{\infty} v^2 \exp\left\{-\frac{m}{2kT}\left[(v_x-u_x)^2+(v_y-u_y)^2+(v_z-u_y)^2\right]\right\} d\vec{v}$$

Substitute $C_i = v_i - u_i$, and $v^2 = C^2 + 2u_iC_i + u^2$

$$\langle v^2 F\rangle = (n_1+n_2)\left(\frac{m}{2\pi kT}\right)^{3/2}\iiint_{\infty}\left(C^2 + \cancel{2u_iC_i}^{0} + u^2\right)\exp\left\{-\frac{m}{2kt}C^2\right\} d\vec{C}$$

(always add integrand)

$$= (n_1+n_2)\left(\frac{m}{2\pi kT}\right)^{3/2}\left(u^2 I_0^3 + 3I_0^2 I_2\right)$$

$$= (n_1+n_2)\left(u^2 + 3\left(\frac{m}{2\pi kT}\right)^{3/2}(2\pi)^{3/2}\left(\frac{kT}{m}\right)^{5/2}\right)$$

$$= (n_1+n_2)\left(u^2 + 3\left(\frac{kT}{m}\right)\right)$$

$$p = \frac{1}{3}\left(m\langle v^2 F\rangle - \rho u^2\right) = \frac{1}{3}\left(3(n_1+n_2)kT + m(n_1+n_2)u^2 - m(n_1+n_2)u^2\right)$$

$$\boxed{p = (n_1+n_2)kT}$$

: $\boxed{T_1 = T_2 = T}$, and $\vec{u}_1 = \vec{u}_2$. if $\vec{u}_1 = \vec{u}_2$, then

$$\vec{u} = \frac{\vec{u}_1 n_1 + \vec{u}_2 n_2}{n_1+n_2} = \frac{\vec{u}_1\cancel{(n_1+n_2)}}{\cancel{(n_1+n_2)}} = \boxed{\vec{u}_1 = \vec{u}_2 = \vec{u}}$$

, where $\vec{u}$ is a new bulk velocity $\vec{u} = (u_x, u_y, u_z)$

2. NDF for spatially uniform gas is... $F(\vec{v}, t=0) = F_0(\vec{v})$

a) $\frac{\partial F}{\partial t} + \vec{v} \cdot \vec{\nabla}_x F + \vec{a} \cdot \vec{\nabla}_{\vec{v}} F = \frac{\delta F}{\delta t}$ (full Boltzmann Equation)

(spatially homogeneous) $\vec{\nabla}_x F = 0$; (no external forces) $\vec{\nabla}_{\vec{v}} F = 0$

$\frac{\partial F}{\partial t} = \frac{\delta F}{\delta t}$ $\xrightarrow[\text{approximation}]{\text{apply BGK}}$ $\boxed{\frac{\partial F}{\partial t} = -\frac{F(\vec{v},t) - M(\vec{v},t)}{\tau}}$

b) given $F(\vec{v},t) - M(\vec{v},t) = \left(F_0(\vec{v}) - M_0(\vec{v})\right) e^{-t/\tau}$, we can take $\frac{\partial F}{\partial t} = -\frac{F(\vec{v},t) - M(\vec{v},t)}{\tau}$ from a), and then substitute in our new relationship for $F(\vec{v},t) - M(\vec{v},t)$

$$\frac{\partial F}{\partial t} = -\frac{\left(F_0(\vec{v}) - M_0(\vec{v})\right) e^{-t/\tau}}{\tau}$$

$$F(\vec{v},t) = \left(F_0(\vec{v}) - M_0(\vec{v})\right)\left[e^{-t/\tau}\right]_0^t + f(\vec{v})$$

($f(\vec{v})$: some arbitrary function of only $\vec{v}$ that could have been zeroed by $\frac{\partial}{\partial t}$.)

$$= \left(F_0(\vec{v}) - M_0(\vec{v})\right)\left[e^{-t/\tau} - 1\right] + f(\vec{v})$$

$$= F_0(\vec{v})\left(e^{-t/\tau} - 1\right) - M_0(\vec{v})\left(e^{-t/\tau} - 1\right) + f(\vec{v})$$

we can then impose: $F(\vec{v},0) = F_0(\vec{v})$

$$F(\vec{v},0) = F_0(\vec{v})\,(1-1) - M_0(\vec{v})\,(1-1) + f(\vec{v})$$

$$= f(\vec{v}) = F_0(\vec{v})$$

$$F(\vec{v},t) = F_0(\vec{v})\left(e^{-t/\tau} - 1\right) - M_0(\vec{v})\left(e^{-t/\tau} - 1\right) + F_0(\vec{v})$$

$$\boxed{F(\vec{v},t) = \left(F_0(\vec{v}) - M_0(\vec{v})\right) e^{-t/\tau} + M_0(\vec{v})}$$

3. comparison of CE and Grad (20-moment) approaches.

a) Approximate forms of the non-equilibrium NDF

In CE: $F = \mathcal{M} \cdot (f^{(0)} + \epsilon^1 f^{(1)} + \epsilon^2 f^{(2)} + \epsilon^3 f^{(3)} + \dots)$

where $\mathcal{M}$ is the Maxwellian distribution, and ϵ is some scaling parameter s.t. $\epsilon \ll 1$

→ this means that F is just a perturbative expansion about $\mathcal{M}$, where the order of the expansion is determined by the allowable order of ϵ preceding terms.

In Grad 20-moment: $\mathcal{F} = \mathcal{M}[1 + \mathcal{P}^{(20)}(C_i)]$, where $\mathcal{P}^{(20)}$ is a 3rd degree polynomial in random velocity C_i

→ $\mathcal{P}^{(20)}(C_i) = E + A_\alpha C_\alpha + \beta_{\alpha\beta} C_\alpha C_\beta + D_{\alpha\beta\gamma} C_\alpha C_\beta C_\gamma$

where the 20 coefficients represented by E (1), A (3), B (6), and D (10) are related by the 20 known moments.

↳ thus, $\mathcal{P}^{(20)}$ actually provides closure by relating the moments to each other

→ the moment constraints are...

↳ $m\langle F^{(20)} \rangle = 0$, $m\langle C_i F^{(20)} \rangle = 0$, $m\langle C_i C_j F^{(20)} \rangle = P_{ij} = p\delta_{ij} - \tau_{ij}$

$m\langle C_i C_j C_k F^{(20)} \rangle = Q_{ijk}$

↳ these constraints relate the macroscopic quantities to the coefficients required to fit the NDF to the maxwellian

b) Macroscopic quantities required to describe the distribution functions

In CE: m, mC_i, and $mC^2/2$ are taken regardless of the order of approximation.

→ higher moments for q_i, τ_{ij} are just used for completing the set of equations, and are not actually computed as their own moment equations.

In Grad 20-moment: $V^{(20)}(V_i) = m\left[1, V_i, V_iV_j, V_iV_jV_k\right]^T$ are considered

→ and hence 20 moments are coupled to the 20 constants in $P^{(20)}$ that form the adjustment to the NDF $F = \mathcal{M} \cdot P^{(20)}$

→ the central moments can also be used

→ here, $P^{(20)}$ provides closure information.

c) generalized transport equations.

In CE: we just have the transport equations for continuity, energy, and momentum.

→ $\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_i} \rho u_i = 0$

→ $\frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j} + \frac{1}{\rho}\frac{\partial p}{\partial x_i} - \frac{1}{\rho}\frac{\partial \tau_{ij}}{\partial x_j} = 0$

→ $\frac{\partial p}{\partial t} + u_i \frac{\partial p}{\partial x_i} + \frac{5}{3} p \frac{\partial u_i}{\partial x_i} + \frac{2}{3}\frac{\partial q_i}{\partial x_i} - \frac{2}{3}\tau_{ij}\frac{\partial u_i}{\partial x_j} = 0$

} for first order. (5 total eqns.)

→ to recover zeroth order, set $q_i = 0$ $\tau_{ij} = 0$ (with more terms popping up at higher orders of perturbative expansion.)

In Grad 20-moment: we apply Maxwell's Eq. of change to all moments, which eventually yields the moment equations.

→ $\frac{\partial}{\partial t}(\rho) + \frac{\partial}{\partial x_i}(\rho u_i) = 0$

→ $\frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j} + \frac{1}{\rho}\frac{\partial P_{ij}}{\partial x_j} = \frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j} + \frac{1}{\rho}\frac{\partial p}{\partial x_i} - \frac{1}{\rho}\frac{\partial \tau_{ij}}{\partial x_j} = 0$

→ $\frac{\partial P_{ij}}{\partial t} + \frac{\partial}{\partial x_k}(u_k P_{ij}) + P_{jk}\frac{\partial u_i}{\partial x_k} + P_{ik}\frac{\partial u_j}{\partial x_k} + \frac{\partial Q_{ijk}}{\partial x_k} = \frac{1}{\tau}\tau_{ij}$

→ $\frac{\partial Q_{ijk}}{\partial t} + \frac{\partial}{\partial x_l}(u_l Q_{ijk}) + Q_{jkl}\frac{\partial u_i}{\partial x_l} + Q_{ikl}\frac{\partial u_j}{\partial x_l} + Q_{ijl}\frac{\partial u_k}{\partial x_l} - \frac{P_{jk}}{\rho}\frac{\partial P_{il}}{\partial x_l} - \frac{P_{ik}}{\rho}\frac{\partial P_{jl}}{\partial x_l} - \frac{P_{ij}}{\rho}\frac{\partial P_{kl}}{\partial x_l} + \frac{\partial R_{ijkl}}{\partial x_l} = -1/\tau\, Q_{ijk}$

where $R_{ijkl} = \frac{1}{\rho}\left[P_{ij}P_{kl} + P_{ik}P_{jl} + P_{il}P_{jk} - \tau_{ij}\tau_{kl} - \tau_{ik}\tau_{jl} - \tau_{il}\tau_{jk}\right]$

(specifically for the BGK collision operator)

→ for other choices of collision operator, this changes.

→ without specifying a collision operator, we can just say the transport equations are ... (for moments provided in b))

$$\frac{\partial}{\partial t}(M_0) + \frac{\partial}{\partial x_i}(u_i M_0) + \frac{\partial}{\partial x_i}\left[\langle C_i V(c_k) F\rangle\right] + \left(\frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j}\right)\left[\left\langle \frac{\partial V}{\partial c_i} F\right\rangle\right] + \frac{\partial u_i}{\partial x_j}\left[\left\langle c_j \frac{\partial V}{\partial c_i} F\right\rangle\right] - \left\langle a_i \frac{\partial V}{\partial c_i} F\right\rangle = \left\langle V(c_k)\frac{\delta F}{\delta t}\right\rangle$$

where ...

$$F = \mathcal{M}(\rho, u_\alpha, P)\left[1 - \frac{\rho}{2P^2}\tau_{\alpha\beta}c_\alpha c_\beta + \frac{\rho^2}{6P^3}\left(Q_{\alpha\beta\gamma}c_\beta c_\gamma - \frac{3P}{\rho}Q_{\alpha\beta\beta}\right)c_\alpha\right]$$

↳ is given by the moment set and reasonable form of the NDF.

→ and there is one such equation for each moment specified in b)

4. a) the Grad 10-moment closure consists of $V^{(10)}(v_i) = m[1, v_i, v_i v_j]$

this model can be obtained from the 20-moment closure by setting $Q_{ijk} = 0$, and by extension $q_i = 0$

this model is called a 10-moment closure because there are a total of 10 velocity moments that must be taken.

$V(v_i) = m, mv_x, mv_y, mv_z, mv_x^2, mv_y^2, mv_z^2, mv_x v_y, mv_x v_z, mv_y v_z$

b) using Grad's method, we have the form

$$F^{(10)}(x_i, c_i, t; \rho, u_\alpha, p, \tau_{\alpha\beta}) = \mathcal{M}(\rho, u_i, P)\left[1 - \frac{\rho}{2p^2}\tau_{\alpha\beta} c_\alpha c_\beta\right]$$

c) the moment equations with $a_i = 0$ are given by...

$$\frac{\partial}{\partial t}(M_0) + \frac{\partial}{\partial x_i}(u_i M_0) + \frac{\partial}{\partial x_i}\left[\langle c_i V_i(c_i) F^{(10)}\rangle\right] + \left(\frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j}\right)\left[\left\langle \frac{\partial V}{\partial c_i} F^{(10)}\right\rangle\right] + \frac{\partial u_i}{\partial x_j}\left[\left\langle c_j \frac{\partial V}{\partial c_i} F^{(10)}\right\rangle\right] = \left\langle V(c_i)\frac{\delta F^{(10)}}{\delta t}\right\rangle$$ Maxwell's EOC.

with $M_0 = \langle V(c_i) F^{(10)}\rangle$ (for the whole set of moments described in a))

begin by stating $\langle V(c_i) F^{(10)}\rangle = \left[1 - \frac{\rho}{2p^2}\tau_{\alpha\beta} c_\alpha c_\beta\right]\langle V(c_i)\mathcal{M}(\rho, u_i, P)\rangle$

we need the following quantities...

$\langle m F^{(10)}\rangle = \rho$

$\langle m c_i F^{(10)}\rangle = 0$

$\langle m c_i c_j F^{(10)}\rangle = \delta_{ij} P - \tau_{ij} = P_{ij}$

$\langle m c_i c_j c_k F^{(10)}\rangle = Q_{ijk}$

$\langle 0\rangle = 0$

$$\frac{\partial c_i c_j}{\partial c_l} = c_i \frac{\partial c_j}{\partial c_l} + c_j \frac{\partial c_i}{\partial c_l} = c_i \delta_{jl} + c_j \delta_{il}$$

then we can also apply the BGK collision operator to get

$$\left\langle V(c_i)\frac{\delta F^{(10)}}{\delta t}\right\rangle = -\frac{1}{\tau}\left[M_0 - \langle V(c_i)\mathcal{M}\rangle\right]$$

so for Maxwell's EOC with $V = m \ldots$

$$\frac{\partial}{\partial t}(\rho) + \frac{\partial}{\partial x_i}(u_i \rho) + \frac{\partial}{\partial x_i}(0) + \left(\frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j}\right)(0) + \frac{\partial u_i}{\partial x_j}(0) = -\frac{1}{\tau}(\rho - \rho)$$

$$\frac{\partial}{\partial t}(\rho) + \frac{\partial}{\partial x_i}(u_i \rho) = 0$$

and with $V = mC_i \ldots$

$$\frac{\partial}{\partial t}(0) + \frac{\partial}{\partial x_i}(u_i (0)) + \frac{\partial}{\partial x_i}[\delta_{ij} P - \tau_{ij}] + \left(\frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j}\right)[\rho] + \frac{\partial u_i}{\partial x_j}[0]$$

$$= -\frac{1}{\tau}[0 - 0]$$

$$\frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j} + \frac{1}{\rho}\frac{\partial}{\partial x_i}[\delta_{ij} P - \tau_{ij}] = 0$$

and with $V = m C_i C_j \ldots$

$$\frac{\partial P_{ij}}{\partial t} + \frac{\partial}{\partial x_k}(u_k P_{ij}) + \frac{\partial}{\partial x_i} Q_{ijk} + \left(\frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j}\right)[0] + \frac{\partial u_l}{\partial x_k}[\delta_{jl} P_{ki} + \delta_{il} P_{kj}]$$

$$= -\frac{1}{\tau}[P_{ij} - p\delta_{ij}] = \frac{1}{\tau}\tau_{ij}$$

$$\frac{\partial P_{ij}}{\partial t} + \frac{\partial}{\partial x_k}(u_k P_{ij}) + \frac{\partial u_i}{\partial x_k}[P_{ki} + P_{kj}] = \frac{1}{\tau}\tau_{ij} \qquad \text{(since } Q_{ijk} = 0 \text{ for 10-moment)}$$

$$\frac{\partial P_{ij}}{\partial t} + \frac{\partial u_k}{\partial x_k} P_{ij} + \frac{\partial P_{ij}}{\partial x_k} u_k + \frac{\partial u_i}{\partial x_k} P_{kj} + \frac{\partial u_j}{\partial x_k} P_{ki} = \frac{1}{\tau}\tau_{ij}$$

mass and momentum are collisional invariants. energy is dissipated through the stress tensor with decay time τ, but isn't fully resolved with Q_{ijk}

d) The 10-moment model doesn't include heat flux. this means it cannot deal with heat transfer, and that is why it is insufficient relative to N.S.

It does possess a valid Euler limit for flows in L.T.E. to recover this form, set $\tau_{ij} = 0$

e) assuming τ is short, we can consider scaled perturbative

solutions for τ_{ij} of the form $\tau_{ij} = \tau_{ij}^{(0)} + \epsilon\,\tau_{ij}^{(1)} + \epsilon^2\tau_{ij}^{(2)} + \dots$ where the scaled moment equation of interest is...

$$\frac{\partial P_{ij}}{\partial t} + \frac{\partial u_k}{\partial x_k}P_{ij} + \frac{\partial P_{ij}}{\partial x_k}u_k + \frac{\partial u_i}{\partial x_k}P_{kj} + \frac{\partial u_j}{\partial x_k}P_{ki} = \frac{1}{\epsilon\tau}\tau_{ij}$$

$$\epsilon\left[\frac{\partial P_{ij}}{\partial t} + \frac{\partial u_k}{\partial x_k}P_{ij} + \frac{\partial P_{ij}}{\partial x_k}u_k + \frac{\partial u_i}{\partial x_k}P_{kj} + \frac{\partial u_j}{\partial x_k}P_{ki}\right] = \frac{1}{\tau}\tau_{ij}$$

$$\tau_{ij} = \tau\epsilon\left[\frac{\partial P_{ij}}{\partial t} + \frac{\partial u_k}{\partial x_k}P_{ij} + \frac{\partial P_{ij}}{\partial x_k}u_k + \frac{\partial u_i}{\partial x_k}P_{kj} + \frac{\partial u_j}{\partial x_k}P_{ki}\right]$$

begin by constructing a zero'th order solution $\tau_{ij} = \tau_{ij}^{(0)}$

$$\tau_{ij} = \tau\epsilon\left[\frac{\partial P_{ij}}{\partial t} + \frac{\partial u_k}{\partial x_k}P_{ij} + \frac{\partial P_{ij}}{\partial x_k}u_k + \frac{\partial u_i}{\partial x_k}P_{kj} + \frac{\partial u_j}{\partial x_k}P_{ki}\right]$$

$$\tau_{ij} = \tau_{ij}^{(0)} = 0$$

now construct a first order solution $\tau_{ij} = \tau_{ij}^{(0)} + \epsilon\tau_{ij}^{(1)} = \epsilon\tau_{ij}^{(1)}$ (0 (from above))

$$\tau_{ij} = \tau\epsilon\left[\frac{\partial P_{ij}}{\partial t} + \frac{\partial u_k}{\partial x_k}P_{ij} + \frac{\partial P_{ij}}{\partial x_k}u_k + \frac{\partial u_i}{\partial x_k}P_{kj} + \frac{\partial u_j}{\partial x_k}P_{ki}\right]$$

Since $P_{ij} = \delta_{ij}p - \tau_{ij}$, and $\tau_{ij} = \epsilon\tau_{ij}^{(1)}$, we can expand to the following form, and then discard the second order terms.

$$\epsilon\tau_{ij}^{(1)} = \tau\epsilon\left[\frac{\partial}{\partial t}(\delta_{ij}p - \tau_{ij}^{(1)}\epsilon) + (\delta_{ij}p - \tau_{ij}^{(1)}\epsilon)\frac{\partial u_k}{\partial x_k} + u_k\frac{\partial}{\partial x_k}(\delta_{ij}p - \tau_{ij}^{(1)}\epsilon) + (\delta_{kj}p - \tau_{ki}^{(1)}\epsilon)\frac{\partial u_i}{\partial x_k} + (\delta_{ij}p - \tau_{kj}^{(1)}\epsilon)\frac{\partial u_i}{\partial x_k}\right]$$

$$\tau_{ij}^{(1)} = \tau\left[\delta_{ij}\frac{\partial}{\partial t}(p) + u_k\delta_{ij}\frac{\partial}{\partial x_k}(p) + \delta_{ij}p\frac{\partial u_k}{\partial x_k} + \delta_{ki}p\frac{\partial u_i}{\partial x_k} + \delta_{kj}p\frac{\partial u_j}{\partial x_k}\right]$$

$$\tau_{ij}^{(1)} = \tau\left[\delta_{ij}\frac{\partial}{\partial t}(p) + u_k\delta_{ij}\frac{\partial}{\partial x_k}(p)\right] + \mu\left[\delta_{ij}\frac{\partial u_k}{\partial x_k} + \delta_{ki}\frac{\partial u_i}{\partial x_k} + \delta_{kj}\frac{\partial u_j}{\partial x_k}\right]$$

where $\mu = \tau p$ (dynamic viscosity), as we can see, this is also not equal to the stress tensor in the N.S. equations.